

Do left-wing or female representatives reduce violence against women? RDD evidence from Brazil*

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Abstract

A growing set of literature explores the effect of the election of female candidates on social outcomes without taking ideology into account. This essay aims to fulfill this gap by examining whether political affiliation contribute—alongside, and cumulatively—to the reduction of Interpersonal Violence against Women (IPVAW). Combining data from the `electionsBR` package and DATA-SUS/SINAN on brazilian municipalities between 2016 and 2025, the article follow a sharp Regression Discontinuity Design (RDD) to explore whether barely winning elections have an impact on the IPVAW rate. The results reinforce Bochenkova et al. (2023) previous findings : gender matters more, while political orientation has limited effects.

Keywords. Interpersonal Violence against Women, Left, Female mayor, Brazil, Regression Discontinuity

*Number of words: 3,500

1 Introduction

A growing body of political sciences, majorly inspired by liberal feminist literature, finds that electing women improves social outcomes such as education spending, public health, while reducing poverty and insecurity (Bochenkova et al., 2023; Clayton & Zetterberg, 2018). However, few studies have empirically digested intersectional materialist feminist critiques. Specifically, the effect of a candidate’s gender in comparison to their political affiliation on Interpersonal Violence against Women (IPVAW) remains largely untested.

This distinction is crucial for real-world policymaking. To effectively reduce violence, we must know whether descriptive and symbolic representation (electing women) or substantive representation (marked by ideology and consent to social order) drives change.

Thus, the paper tests the following inquiry:

RQ: Does a candidate’s gender or party orientation—or both simultaneously—equally reduce Interpersonal Violence against Women?

This essay contributes to the feminist debate by quantitatively testing the intersectional materialist framework using real-world Brazilian election data. I combine a sharp RDD with IPVAW data, using the margin of victory of the candidate as the running variable. In the prolongation of Bochenkova et al. (2023), it is found that female representatives from various political affiliations were more effective in reducing the IPVAW rate than left-wing, male mayors.

2 Literature review

2.1 Theoretical family

The theoretical framework for analysing women’s political representation is based heavily on Pitkin’s foundational distinction between descriptive, substantive, and symbolic representation (Pitkin, 1967). Descriptive representation refers to the physical presence of representatives who share demographic

characteristics—such as gender—with their constituents. Substantive representation, by contrast, concerns the ideological alignment between representatives and those they represent, regardless of shared identity. Symbolic representation refers to the broader cultural and affective dimensions: the ways in which the mere presence of certain representatives shapes citizens' beliefs about legitimacy, belonging, and the responsiveness of political institutions (Göhler et al., 2013; Pitkin, 1967).

An important body of evidence suggests that descriptive representation matters. Rosenberger (2017) has documented a "role-model effect" whereby visible female leaders inspire political engagement among young women and girls. Recent work on symbolic representation have emphasised that symbols carry both cognitive and affective dimensions (Göhler et al., 2013), meaning that the mere presence of women in positions of power transforms how citizens perceive political institutions and their own place within them.

2.2 The Policy Consequences of Female Political Leadership

Beyond conceptual debates, a large quantitative literature has examined whether electing women produces measurable changes in policy outcomes. Using a Difference-in-Differences approach, Clayton and Zetterberg (2018) have demonstrated that gender quotas increase the share of women in parliaments, increasing in turn public goods provision—particularly in education and health—while reducing military expenditures.

In the Indian context, Chattopadhyay and Duflo (2004) use a natural experiment of village councils to show that female leaders invest more in infrastructure and services that reflect the stated priorities of women, such as drinking water and sanitation. Collectively, this literature establishes that descriptive representation of women often causes different spending priorities and social outcomes compared to male-dominated governance.

2.3 Female Leadership and Interpersonal Violence: Causal Evidences

In the US context, Wen (2025) found that electing female mayors increases rape reports to police by 30 to 37%. Rather than indicating an increase in actual violence, the author interprets this as a "role-model" or "reporting" effect: women feel more empowered to report crimes when led by female officials.

Most importantly for this article, Bochenkova et al. (2023) provide evidence that descriptive representation of women reduces violence against women. Using an RDD on Brazilian municipal elections from 2004 to 2016, they find that electing a female mayor reduces feminicides by approximately 50% over a four-year term. These findings suggest that female political leadership reduces IPVAW—whether through direct policy changes, improved surveillance, or increased reporting.

However, this literature treats women as a homogeneous category with stable policy preferences. Three caveats warrant attention. First, the causal mechanism remains unclear: feminicides often occur when women leave or are about to leave an abusive relationship, a channel largely unaffected by mayoral policy. Second, Bochenkova et al. (2023) employ bandwidths as large as 15 percentage points in vote margin—a threshold that stretches the definition of a "close election" and potentially violates the RDD identifying assumption. Third, and most critically for this study, the authors do not examine whether political orientation dominates the effect of female leadership.

2.4 The Intersectional and Materialist Critique: A Missing Dimension

The treatment of women as a unified political subject has been challenged from two related theoretical traditions. Intersectional feminism, pioneered by Crenshaw (1994), argues that gender cannot be isolated from race, class, and other axes of identity. As Crenshaw (1994) demonstrates, racialized

women face a unique representational dilemma. They cannot be adequately represented by liberal, white-dominated feminist movements, which neglect the specific oppressions of racialized women. Yet they also cannot fully align with anti-racist movements, which tend to invisibilize violence against women within their communities to avoid reinforcing racist stereotypes of the "savage". The implication for political representation is profound: descriptive representation of women may primarily benefit privileged women while leaving others unaffected—or even actively harmed.

Materialist feminism, particularly the Francophone tradition articulated by Guillaumin (1978) and Delphy and Leonard (1980), offers a complementary critique. This perspective demonstrates that "sex" constitutes a structural social relation akin to class: gender relations are relations of production. A female politician's position in the social hierarchy—including her class background, her party's relationship to capital, and her willingness to challenge established order—profoundly shapes her policy preferences. They argue that liberal feminist approaches, by treating women as a universal category, obscure how wealthy conservative women may actively oppose redistributive policies that would reduce violence for poor women.

The materialist critique predicts that electing a woman matters only insofar as that woman challenges the structural inequalities producing violence. A female politician from a conservative party, allied with established economic interests and supportive of carceral rather than social welfare approaches, may produce no reduction in violence—or may even exacerbate it by defunding social services. Conversely, a male politician from a left-wing party committed to transformative social policy could plausibly reduce violence more than a conservative female counterpart.

Despite the theoretical importance of these critiques, empirical tests remain rare. Most studies estimate the average treatment effects of female candidates, implicitly assuming homogeneity. To date, no studies have tested whether the effect of female leadership on IPVAW was more important than political orientation.

3 Theoretical Framework

This section develops the theoretical argument that the effect of the gender and the ideology on interpersonal violence against women (IPVAW). First, I conceptualize the key variables. Second, I specify the causal mechanisms linking mayoral characteristics to IPVAW. Third, I derive testable hypotheses. Fourth, I define scope conditions.

3.1 Conceptualization

3.1.1 Female Mayor

This study operationalizes female mayor as a binary indicator based on electoral registry records. While the theoretical mechanism may involve gender-based perceptions (socially constructed roles), available data only record biological sex. I therefore use sex as a proxy for gender.

3.1.2 Left-Wing Party

Following the Brazilian political science literature (Kinzo, 2004; Nogueira, 2019), a party is classified as left-wing if it belongs to the ideological family that has historically prioritized redistributive social policies, labor rights, and gender-based violence prevention. The list is available in the Appendix 7.

3.1.3 Interpersonal Violence against Women (IPVAW)

Interpersonal violence against women (IPVAW) conceptualization flows from the United Nations Declaration on the Elimination of Violence against Women (CEDAW, 1995). It is considered as any act of violence perpetrated by a non-state individual against a woman, involving at least one of three dimensions: (1) physical violence (e.g., battery, weapons), (2) psychological violence (e.g., threats, humiliation), or (3) sexual violence (e.g., assault, coercion).

3.2 Causal Mechanisms

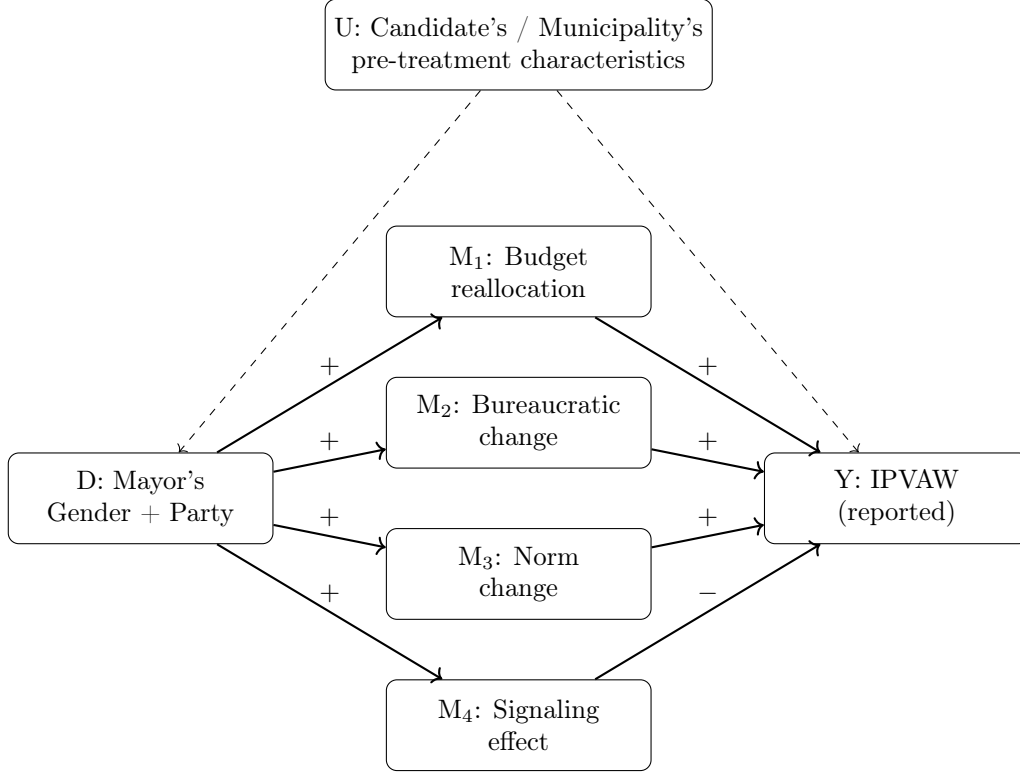
Ideological foundations. Drawing on the manifesto project data for Brazilian municipal elections, Tarouco and Madeira (2015) shows that left-wing parties are significantly more likely to include explicit commitments to violence against women prevention, to propose dedicated women’s secretariats, and to integrate racialized minorities and women into public institutions. Conversely, non-left parties tend to frame violence as a criminal justice issue rather than a structural one, and rarely include gender-based violence prevention as a central campaign pledge. This programmatic divergence suggests that left-wing mayors, once elected, face coalitional incentives and electoral accountability pressures that non-left mayors do not, making them more likely to translate descriptive representation into substantive policy change.¹ Thus, four causal mechanisms are theoretically plausible, from mayoral characteristics to IPVAW (presented in figure 1).

First pathway: Budget reallocation. A mayor controls municipal budget allocation. Female mayors in Brazil have been shown to prioritize health, education, and social protection—sectors that include VAW prevention (Bochenkova et al., 2023). However, this priority is not automatic: substantive representation could matter since left-wing mayors may face coalitional pressure from feminist movements and labor unions to fund shelters, hotlines, and prevention programs. Conversely, non-left mayors could face pressure from coalitions favoring policing and carceral solutions. Thus, a left-wing male mayor is predicted to reallocate more budget toward structural VAW prevention than a non-left female mayor.

Second pathway: Bureaucratic restructuring. Mayors appoint secretaries and shape administrative priorities. Female mayors are more likely to appoint women to security and social service positions (Wen, 2025). These

¹Importantly, I do not claim left parties represent *all* women as a homogeneous group—which would contradict the intersectional premise. Rather, left parties represent the *material conditions* under which the *majority* of potential victims of IPVAW live. Conversely, I am not claiming that electing a woman will always lead to perfect descriptive representation, since this also encompasses race and class. This remains an average effect.

Figure 1: DAG: Causal pathways from mayor characteristics to IPVAW



appointees, in turn, influence police training, victim support protocols, and inter-agency coordination. Left-wing mayors may be specifically incentivised to create dedicated women's secretariats and to implement progressive protocols. Non-left mayors may maintain or even weaken existing protective structures.

Third pathway: Norm change. A mayor's public stance shapes social norms about violence acceptability. When a female mayor publicly prioritizes VAW prevention, she signals that violence is unacceptable and that victims will be believed. Left-wing mayors, if embedded in movements that already challenge to a certain extent patriarchal norms, are more likely to make such public commitments. Non-left mayors may avoid the issue or frame it as a matter of individual criminality rather than structural change.

Fourth pathway: Reporting signal. The presence of a female mayor affects victims’ willingness to report violence. Victims who see a woman in power may perceive institutions as less hostile (Wen, 2025). This reporting effect increases recorded IPVAW even if actual violence remains constant. Left-wing mayors may amplify this effect through visible public campaigns. The crucial distinction lies in whether increased reporting is accompanied by effective intervention.

While these four pathways are theoretically plausible, the RDD design identifies the reduced-form effect of electing a female mayor on reported IPVAW, not the mechanisms themselves. Disentangling these pathways requires data on budgets, appointments, and reporting behavior—a potential avenue for future research.

3.3 Hypothesis

The theoretical framework yields three nested hypotheses, moving from simpler to more complex predictions.

Party ideology main effect. If the intersectional-materialist mechanism operates through coalitional incentives and substantially represents the interests of a broad set of women, then left-wing mayors (male or female) should produce larger reductions in IPVAW than non-left mayors.

H1: Municipalities with a left-wing mayor experience lower reported IPVAW than those with a non-left mayor, regardless of the mayor’s gender.

Alternative: Gender effect. If gendered descriptive representation matters, then:

H2: Municipalities with a female mayor experience lower reported IPVAW than those with a male mayor, regardless of the mayor’s political orientation.

Comparative test. The intersectional-materialist framework implies that substantive representation could matter at least as much as gendered descriptive representation alone. Thus, I test:

H3: Municipalities with a left-wing male mayor experience lower reported IPVAW than those with a non-left female mayor.

3.4 Scope Conditions

The theoretical mechanism operates only within specific boundaries. First, the theory applies only to *close elections* where the margin of victory is sufficiently narrow to guaranty continuity at the cut-off. In landslides, the RDD identifying assumption fails because treated and control units differ in pre-treatment characteristics. The continuity assumption is given by:

$$\lim_{x \rightarrow c_0^-} \mathbb{E}[Y_i(0) \mid MV_i = x] = \lim_{x \rightarrow c_0^+} \mathbb{E}[Y_i(0) \mid MV_i = x] \quad (1)$$

2

Second, the theory applies to Brazilian municipalities with strong mayoral systems (i.e., mayor has budget authority and administrative appointment power). It does not necessarily generalize to parliamentary systems, council-manager governments, or non-Brazilian contexts where mayoral powers differ.

Third, the theory applies to the period following the mayors' entry into office. This article focuses on the short-term effect, measured one year after the election. While one might expect that mechanisms such as budgetary reallocation or bureaucratic change require more time to materialize, a one-year window remains informative for several reasons. It captures

²While the continuity of potential outcomes is fundamentally untestable, a necessary (but not sufficient) condition is that pre-treatment covariates (X_i) exhibit no discontinuity at the cutoff:

$$\lim_{x \rightarrow c_0^-} \mathbb{E}[X_i \mid MV_i = x, D_i = 0] = \lim_{x \rightarrow c_0^+} \mathbb{E}[X_i \mid MV_i = x, D_i = 1] \quad (2)$$

where $D_i = 1$ indicates the mayor of interest won (treatment) and $D_i = 0$ indicates they lost (control). This balance condition is empirically testable and provides supporting evidence for the identification strategy. Balance tests are available in the Appendices 7

rapid policy adjustments—such as signaling effects or early administrative reorientations—that may precede more structural changes. Also, testing the immediate aftermath of the election helps isolate the direct influence of the mayor’s characteristics from longer-term confounding factors. In addition, it provides a conservative benchmark: if effects are already visible within one year, this strengthens the case for a causal interpretation.³

Finally, the theory applies to *reported IPVAW* as measured by DATASUS/SINAN. It does not directly speak to unreported violence. Reporting effects (Pathway 4) may produce countervailing pressures that require separate identification.

4 Method and Data

This study investigates the link between mayoral characteristics and the rate of Interpersonal Violence Against Women (IPVAW) across Brazilian municipalities. Brazil offers an ideal case for several reasons. First, the country experiences extreme levels of violence against women: a woman is killed every two hours and assaulted every fifteen seconds on average (Cerqueira & Bueno, 2020). Second, Brazil’s large number of municipalities (5,570) provides substantial statistical power and variation in both electoral outcomes and violence rates. Third, high-quality administrative data on both municipal elections (via the `electionsBR` package) and reported violence (via DATASUS/SINAN) are publicly available and replicable. Finally, this design allows a direct assessment of previous findings from (Bochenkova et al., 2023), extending it by testing the additional role of ideology.

4.1 Sample, Population, and Time Period

The unit of analysis is municipality-year. Brazil has 5,570 municipalities, but the analytic sample is restricted to those with closely contested elections, defined by a margin of victory below a chosen bandwidth. The final dataset

³Note that an alternative tests on subsamples have been conducted four years after the elections like in the original article (Bochenkova et al., 2023), but the results remain very stable.

comprising all the closest races contains 6,413 observations. The descriptive statistics are displayed in Table 1. The data are time-series cross-sectional for the period 2016-2025.⁴

4.2 Identification Strategy

Estimating the causal effect of elections is challenging because an election is not random. A Simple Difference in Outcome (SDO) would suffer from selection bias and heterogeneous treatment effects. Therefore, a sharp Regression Discontinuity Design (RDD) is applied to a sample of tight mixed-characteristics electoral races to make the treatment (election) as good as random.

4.3 Dependent Variable

The dependent variable is the rate of reported Interpersonal Violence Against Women per Brazilian municipality (`ipvaw_rate`). Data on IPVAW come from DATASUS/SINAN⁵. The dataset provides the number of violence occurrences against women that are not self-inflicted. This number is divided by the municipality’s population size to obtain a normalized rate per 100,000 inhabitants. Population data are available from the Brazilian Institute of Geography and Statistics (IBGE)⁶.

SINAN captures reported cases from health facilities, which may under-report due to stigma or fear. However, it is the most comprehensive health-based violence registry in Brazil.

4.4 Independent variables

Information on candidate characteristics is accessible in R using the `electionsBR` package⁷. The vote share, sex, and party affiliation are recorded for each can-

⁴The municipal elections took place in 2016, 2020, and 2024, with the new mayor taking office on 1 January the following year. More information on the institutional background in the Appendices 7.

⁵Available at: <http://tabnet.datasus.gov.br/cgi/tabcgi.exe?sinannet/cnv/violebr.def>

⁶Available at: <https://www.ibge.gov.br/en/statistics/social/population.html>

⁷Available at: <https://electionsbr.com/novo/>

didate. The `female` variable is recoded into a binary variable where female candidate = 1.

Party ideology is coded based on the classification system developed by Kinzo (2004) and updated by Nogueira (2019) for municipal elections. The binary indicator `left` equals 1 for the parties listed in the Appendices 7.

4.5 Control variables

Additional candidate-level information from the `electionsBR` package includes education level (`education`) and year of birth (used to derive `age`). The size of the population is also controlled for at the municipality-level (`population`, in logarithm). These allow testing whether elected female or left-wing mayors are comparable to non-elected ones on observable pre-treatment characteristics. However, no information on the race or income of the candidates is available.

4.6 Descriptive statistics

Table 1: Descriptive statistics

Statistic	N	Mean	St. Dev.	Min	Max
IPVAW Rate (per 100k)	6,162	86.851	132.737	0.000	1,473.077
Margin of Victory (pp)	6,413	-0.292	7.735	-14.998	14.998
Female Mayor (0/1)	6,413	0.079	0.270	0	1
Left-wing Mayor (0/1)	6,413	0.194	0.395	0	1
Age (years)	6,413	50.269	10.675	19	95
Education Level (1-7)	6,413	5.376	1.285	2	7
Population	6,162	9.227	1.028	6.725	16.254

4.7 Model specification

Three subsamples are derived, reflecting the comparisons of interest (presented in Table 2). These three specifications map directly onto the theoretical alternatives presented in Section 3.

Table 2: RDD settings.

Model name	Treatment definition
Substantive representation (L)	left ($L = 1$) vs non-left ($L = 0$)
Descriptive / Symbolic representation (F)	female ($F = 1$) vs male ($F = 0$)
Comparison ($L \times F$)	left male ($L = 1 \times F = 0$) vs non-left female ($L = 0 \times F = 1$)

The effect of municipality i electing a mayor with specific characteristics EC is given by:

$$Y_{it} = \alpha + \beta EC_{it} + f(MV_{it}) + \gamma X_{it} + \epsilon_{it} \quad (3)$$

where Y_{it} denotes the rate of IPVAW in municipality i in year $t \in \{2017, 2021, 2025\}$ (one year after the election, once the mayor has taken office). β is the coefficient of interest, measuring the effect of electing a particular type of mayor. EC_i is the treatment dummy, which equals L_i (left-wing), F_i (female), or $L_i \times F_i$ (interaction) depending on the specification. It captures tight victories by mayors with these characteristics, following the subsamples defined in Table 2. The running variable is MV_i , the margin of victory recentered around the cutoff c_0 , defined as the difference in vote share between the two most-voted candidates. X_i comprises a set of pre-treatment covariates for robustness.

Formally, the estimand of interest is the Local Average Treatment Effect (LATE) at the cutoff:

$$\tau_{SRD} = \lim_{x \rightarrow 0^+} E[Y_i(1) | MV_i = x] - \lim_{x \rightarrow 0^-} E[Y_i(0) | MV_i = x] \quad (4)$$

To test the hypothesis that left-wing candidates reduce the IPVAW rate more than non-left candidates:

$$H_1 : E[Y | L = 1] - E[Y | L = 0] < 0 \quad (5)$$

To test H_2 (female vs male):

$$H_2 : E[Y | F = 1] - E[Y | F = 0] < 0 \quad (6)$$

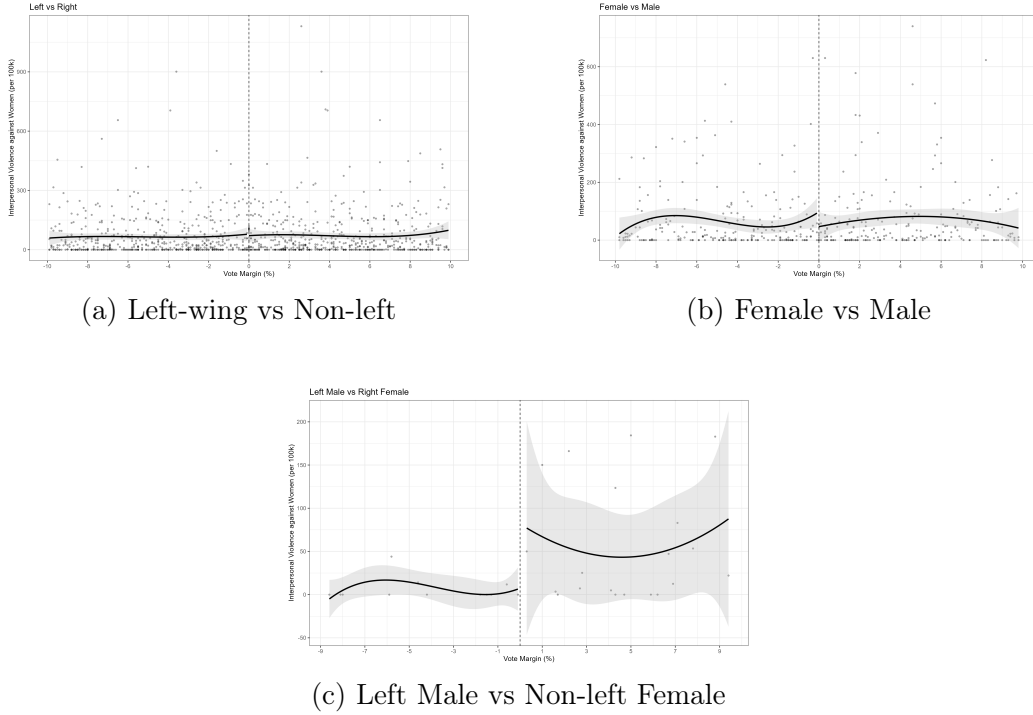
To test H_3 (Comparison: left male vs non-left female):

$$H_3 : E[Y \mid L = 1, F = 0] - E[Y \mid L = 0, F = 1] < 0 \quad (7)$$

5 Results

Table 3 presents the sharp RDD estimates for three specifications testing different theoretical predictions. Figure 2 displays the corresponding discontinuity plots.

Figure 2: RDD estimates for different treatment specifications



5.1 Substantive Representation: Left-wing vs Non-left Mayors

Panel A of Table 3 tests H1: that left-wing mayors (regardless of gender) reduce IPVAW more than non-left mayors. The conventional estimate shows

a negative coefficient (-6.96), indicating lower violence rates under left-wing mayors, but the effect is imprecisely estimated ($p = 0.976$) with confidence intervals largely crossing zero. The bias-corrected and robust estimates actually reverse sign (19.15 and 25.29 respectively), suggesting no systematic discontinuity at the threshold.

These null results indicate that *party ideology alone* does not produce a detectable effect on reported IPVAV in close elections. While the theoretical mechanism predicted substantive representation would matter, the empirical evidence does not support H1 at conventional significance levels.

5.2 Descriptive Representation: Female vs Male Mayors

Panel B tests H2 (the liberal feminist prediction) and reveals a different pattern. The conventional estimate shows that electing a female mayor reduces IPVAV by approximately 60 per 100,000 inhabitants relative to male mayors. The bias-corrected estimate is larger (≈ -80). Although standard errors remain substantial due to the limited sample of close elections involving female candidates ($N_{eff} = 444$ for conventional, dropping to 281 for bias-corrected), the *direction and magnitude* are consistent with previous findings (Bochenkova et al., 2023).

Importantly, the effect is substantially larger than the left-wing coefficient from Panel A (-60.34 vs -6.96), providing preliminary intuition that descriptive representation may matter more than substantive representation alone.

Table 3: RDD Estimates by Treatment Specification

Model	Estimate	95% CI	p-value	Eff. N	Bandwidth
<i>Panel A: Left-wing vs Non-left (N municipalities = 1,040)</i>					
Baseline (b)	-6.96	[-59.70, 57.90]	0.976	1,040	11.00
Bias-corrected (mserd)	19.15	[-50.69, 100.78]	0.517	511	4.31
Robust (cerrd)	25.29	[-56.28, 109.79]	0.528	361	2.93
<i>Panel B: Female vs Male (N municipalities = 444)</i>					
Baseline (b)	-60.34	[-217.44, 58.91]	0.261	444	11.00
Bias-corrected (mserd)	-82.80	[-256.52, 86.77]	0.333	281	5.52
Robust (cerrd)	-81.42	[-263.59, 99.62]	0.376	208	3.93
<i>Panel C: Left Male vs Non-Left Female (N municipalities = 32)</i>					
Baseline (b)	68.52	[-24.80, 163.14]	0.149	32	11.00
Bias-corrected (mserd)	12.35	[-323.40, 336.63]	0.969	12	3.96
Robust (cerrd)	-10.20	[-350.11, 318.72]	0.927	12	3.27

Note: Estimates from sharp RDD with triangular kernel. Baseline (b) uses conventional standard errors; Bias-corrected (mserd) and Robust (cerrd) use MSE-optimal bandwidth selection. CI = confidence interval.

5.3 Comparative Test: Left Male vs Non-left Female

Panel C directly adjudicates between the materialist-intersectional and liberal feminist frameworks. The conventional estimate shows a positive coefficient (68.52), indicating that municipalities with left male mayors have higher IPVAW rates than those with non-left female mayors—contradicting the expectation that left-wing leadership reduces violence.

However, the results reveals two caveats. First, the effective sample size is critically small ($N_{eff} = 32$ for conventional, dropping to just 12 for bias-corrected), explaining the very high p -values. Second, the robust estimate actually reverses sign (-10.20), suggesting extreme sensitivity to bandwidth choice.

5.4 Summary of Findings

Three findings emerge. First, left-wing ideology alone (Panel A) shows no detectable effect on IPVAW. Second, female gender alone (Panel B) shows a substantively large but imprecisely estimated negative effect, consistent with descriptive representation mattering. Thirdly, while the direct comparative test (Panel C) is severely underpowered due to the small sample size, it suggests that women tend to outperform left-wing male mayors, regardless of their political orientation. This contradicts H3.

6 Discussion

6.1 Interpretation of Null and Imprecise Findings

The null result for left-wing ideology (Panel A) does not necessarily indicate that ideology is irrelevant. Rather, it could suggest that any ideology-driven effect is either (a) too small to detect with available statistical power, (b) operates through mechanisms not captured by reported IPVAW rates, (c) motivate the IPVAW report (consistent with Wen (2025)), or (d) necessitate more categories than a simple dummy variable for left versus non-left.

The imprecise but substantively large estimate for female mayors (Panel B) aligns with Bochenkova et al. (2023) while failing to reach conventional significance thresholds.

6.2 The Comparative Test: What Can We Learn from Panel C?

The comparison of left male vs non-left female mayors (Panel C) was proposed as a substitute for the infeasible left female vs non-left female comparison. Theoretically, this test is valid because it isolates the same underlying contrast: does substantive representation (left ideology) or descriptive representation (female gender) matter more?

The statistical fragility of this estimate—the robust specification reverses sign, confidence intervals are enormous, and effective N drops to 12, precluding any definitive conclusion. The data cannot adjudicate between these theoretical alternatives.

6.3 Limitations

Several limitations warrant mention. First, the dependent variable captures *reported* IPVAW, which may reflect reporting behavior rather than actual violence. Second, bandwidth selection for Panel C is unstable given the tiny sample. Third, balance checks reveal minor violations of the continuity assumption: education differs across female treatment and control ($p = 0.04$), and lagged IPVAW differs for the left-male comparison ($p = 0.004$), though these reflect either substantively small differences or small-sample instability.

7 Conclusion

This article tested whether the effect of female mayors on violence against women is conditional on party ideology, or whether descriptive representation matters regardless of political affiliation. The empirical strategy employed a sharp Regression Discontinuity Design using Brazilian municipal election

data (2016–2025), extending the influential work of Bochenkova et al. (2023) by incorporating the intersectional-materialist critique.

Three findings emerge. First, left-wing ideology alone shows no detectable effect on reported IPVAW. Second, electing a female mayor is associated with a substantively large reduction in violence—approximately 60 per 100,000 inhabitants—though the estimate is imprecise due to the limited number of close elections involving female candidates. Third, the critical comparative test between left-wing male and non-left female mayors suffers from severe attrition ($N_{eff} = 32$), precluding definitive conclusions about whether substantive or descriptive representation drives change.

The available evidence offers partial support for descriptive representation, reinforcing Bochenkova et al. (2023)’s core finding. Female mayors appear to reduce violence against women more effectively than left-wing male mayors in the short run, suggesting that gender matters at least as much as ideology. However, this conclusion remains tentative given the imprecise estimates and the inability to rule out alternative explanations, including differential reporting effects (Pathway 4).

Future research should seek to identify which of the four causal channels matters most: the reallocation of municipal budgets toward social services and violence prevention programs; bureaucratic changes, such as the appointment of women to security positions and the creation of dedicated women’s secretariats; norm change, whereby female leaders shift public perceptions of violence acceptability; or the signaling effect, whereby victims feel more empowered to report violence when led by female officials. Disentangling these mechanisms would require data on municipal spending, administrative appointments, and victim reporting behavior—a promising avenue for future work.

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Appendices

Data availability and Replication code

Available at: <https://github.com/SimonLabarthe/TIRDS>

Institutional Background: Local politics

Brazil is a federal presidential republic. The political and administrative organization is a pyramid structured going to the federal government, then 27 federal units, and finally 5,571 municipalities.

Municipalities comprise a mayor (*prefeito/prefeita*) and a legislative body (*câmara municipal*). Each municipality has the constitutional power to approve its own laws, collecting taxes, and receiving funds from upper levels. These local governments are responsible for providing several local public goods (e.g., health care, primary education, municipal infrastructure). In this setting, the mayor plays a key role in defining the expenditure programs, while police activity and law enforcement is under the control of the upper levels. They have the capacity to maintain a municipal police force, pass laws on any matter that do not contradict either the state or the national constitutions, and to create symbols for itself.

In each municipality, an autonomous local government directly elected by the population every four years is designated. Mayors are elected in a one-round election in municipalities with less than 200,000 registered voters, while a run-off may occur in cities with more than 200,000 voters, when no mayoral candidate achieves at least 50% of the votes in the first round.

List of Left parties in Brazil

The variable `left` is coded 1 for parties classified as left-wing or center-left, and 0 otherwise. Table 4 presents the complete classification following Kinzo (2004) and Tarouco and Madeira (2015).

Additional checks

Table 4: Classification of Brazilian Political Parties

Code	Parties
<code>left = 1</code>	PT, PCdoB, PCB, PSB, PSOL, PDT, REDE, PV, PSTU, PCO, UP, PPL, PHS
<code>left = 0</code>	All others (MDB, PSD, UNIÃO, PL, PP, etc.)

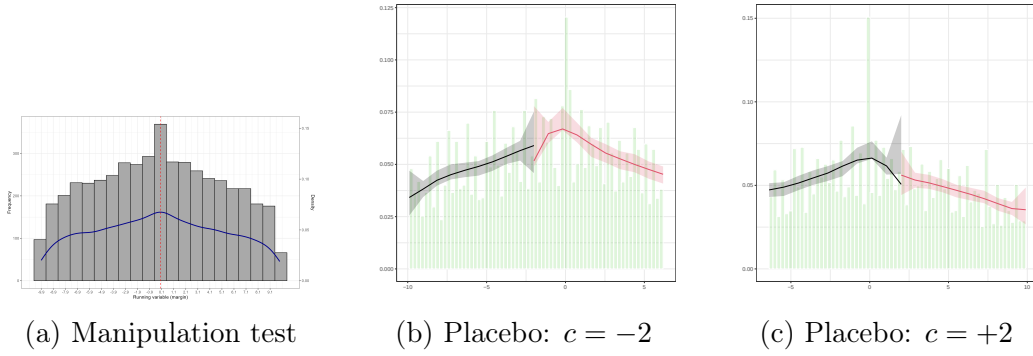


Figure 3: Manipulation and placebo density tests

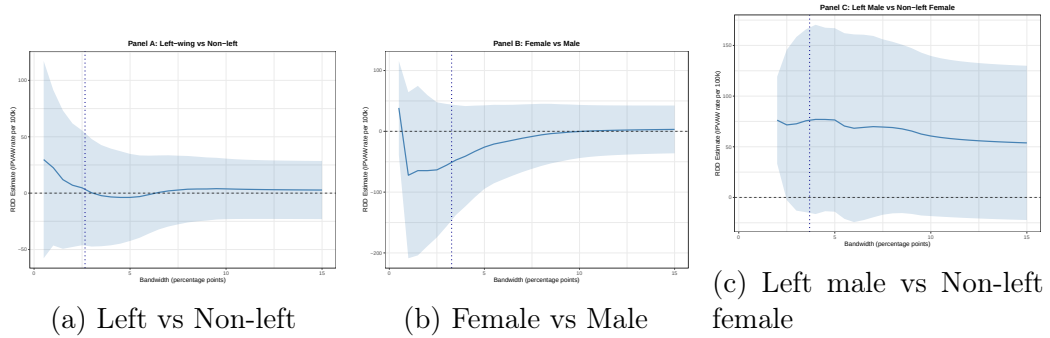


Figure 4: Bandwidth sensitivity checks

Balance checks

Table 5: Balance Test - left

	Control Mean	Control N	Treatment Mean	Treatment N	Difference	P-value
lag_ipvaw_rate	67.14	523	72.10	517	4.97	0.47
age	50.26	549	49.21	545	-1.05	0.09
education	5.28	549	5.31	545	0.02	0.76
population	9.19	523	9.20	517	0.01	0.89

Table 6: Balance Test - female

	Control Mean	Control N	Treatment Mean	Treatment N	Difference	P-value
lag_ipvaw_rate	61.82	227	69.51	217	7.69	0.47
age	49.40	239	48.53	225	-0.87	0.36
education	5.49	239	5.70	225	0.21	0.04
population	9.29	227	9.31	217	0.01	0.87

Table 7: Balance Test - left_male

	Control Mean	Control N	Treatment Mean	Treatment N	Difference	P-value
lag_ipvaw_rate	5.83	12	55.79	20	49.96	0.004
age	47.50	12	45.85	20	-1.65	0.57
education	4.75	12	5.50	20	0.75	0.16
population	8.88	12	9.40	20	0.52	0.08